

LiTDI: A Highly Efficient Additive for Electrolyte Stabilization in Lithium-Ion Batteries

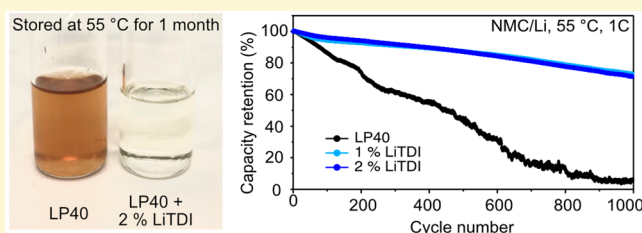
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S Supporting Information

ABSTRACT: The poor stability of LiPF₆-based electrolytes has always been a bottleneck for conventional lithium-ion batteries. The presence of inevitable trace amounts of moisture and the operation of batteries at elevated temperatures are particularly detrimental to electrolyte stability. Here, lithium 2-trifluoromethyl-4,5-dicyanoimidazole (LiTDI) is investigated as a moisture-scavenging electrolyte additive and can sufficiently suppress the hydrolysis of LiPF₆. With 2 wt % LiTDI, no LiPF₆ degradation can be detected after storage for 35 days, even though the water level in the electrolyte is enriched by 2000 ppm. An improved thermal stability is also obtained by employing the LiTDI additive, and the moisture-scavenging mechanism is discussed. The beneficial effects of the LiTDI additive on battery performance are demonstrated by the enhanced capacity retention of both the LiNi_{1/3}Mn_{1/3}Co_{1/3}O₂ (NMC)/Li and NMC/graphite cells at 55 °C. In particular, the increase in cell voltage hysteresis is greatly hindered when LiTDI is presented in the electrolyte. Further development of the LiTDI additive may allow the improvement of elevated-temperature batteries, as well as energy savings by reducing the amount of effort necessary for dehydration of battery components.

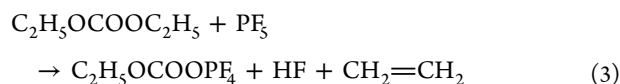
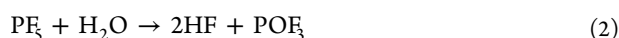


INTRODUCTION

The performance of lithium-ion batteries has been much improved over the past two decades, and the technology is dominating the market of power supplies for portable electronics.^{1–3} In today's state-of-the-art commercial lithium-ion batteries, the most widely used electrolytes are non-aqueous liquid electrolytes, which commonly consist of the LiPF₆ salt and low-molecular weight carbonate solvents as well as various small amounts of functional additives.^{4,5} Although a large number of other lithium salts have been developed as promising alternatives, LiPF₆ still holds the dominating position in commercial liquid electrolytes because of its well-balanced properties, and the situation is unlikely to change for the foreseeable future.^{4,6–8} However, the most severe problems with LiPF₆, which are its high reactivity toward moisture and poor thermal stability, remain unsolved. These issues are primarily attributed to the equilibrium decomposition reaction of LiPF₆, as described in eq 1:^{8–11}



The P–F bond in LiPF₆ and PF₅ is rather labile toward hydrolysis by inevitable trace amounts of moisture in batteries. Besides, as a strong Lewis acid, PF₅ can also initiate reactions with carbonate solvents and cause further electrolyte degradation.^{10,12,13} PF₅ reacting with H₂O and diethyl carbonate (DEC) are illustrated as examples in eqs 2 and 3:^{9,13}



The generated decomposition product, HF, is a serious health hazard because of its high toxicity and highly reactive toward both active and inactive materials in the electrode.^{14,15} Moreover, an increase in temperature further accelerates the decomposition reaction of LiPF₆ and consequently promotes subsequent parasitic reactions. This is also the primary reason for faster aging of lithium-ion batteries at elevated temperatures, as compared to that at room temperature.^{16–19} Until today, there have been substantially fewer electrolyte additives developed that can successfully improve the stability of LiPF₆ compared to those of other types of additive, such as solid electrolyte interphase (SEI)-forming agents.^{4,5,20} Only a few mild Lewis basic compounds as electrolyte additives have been explored with the aim of suppressing PF₅ hydrolysis by formation of more stable complexes.^{21,22} Examples of such compounds are tris-2,2,2-trifluoroethyl phosphite (TTFP), hexamethylphosphoramide (HMPA), and pyridine.^{21,22}

Extensive research has, instead, been conducted on exploring alternative lithium salts, which are chemically and electrochemically more stable than LiPF₆. So far, however, no salts can

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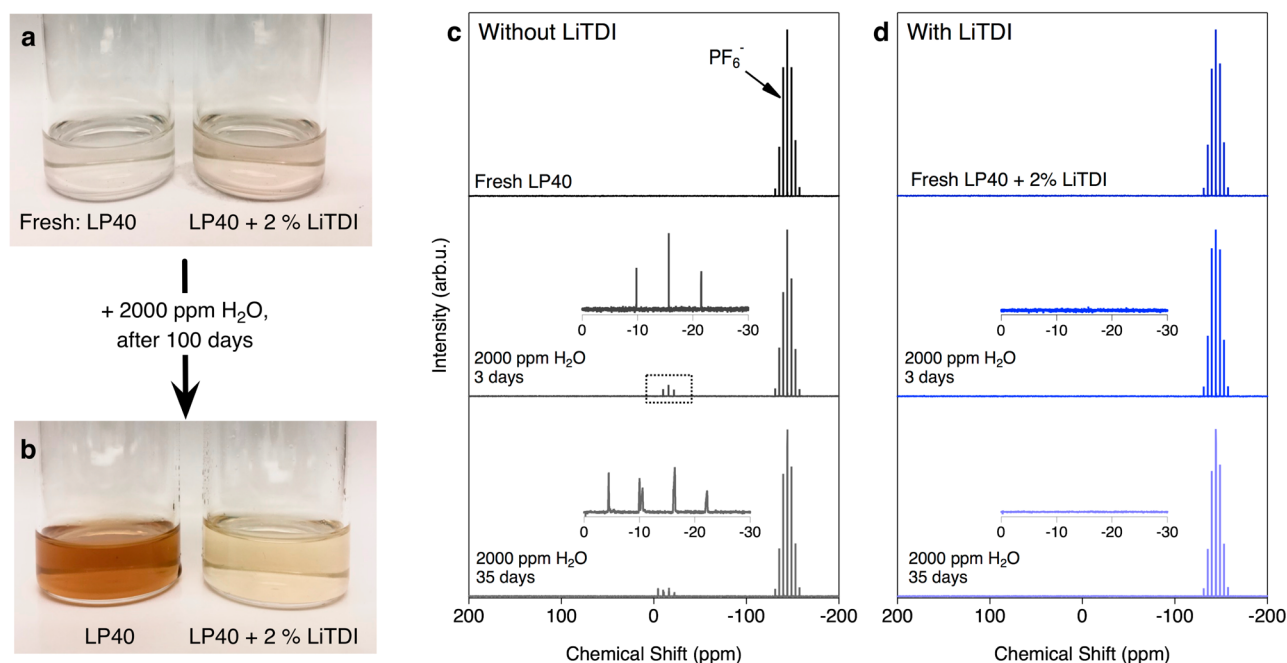


Figure 1. Stability of water-enriched electrolytes. (a) Image of the fresh LP40 (left) and LP40/2% LiTDI (right) electrolytes. (b) Image of LP40 with 2000 ppm of H₂O (left) and LP40 with 2% LiTDI and 2000 ppm of H₂O (right) after storage for 100 days. (c) ³¹P NMR spectra of fresh LP40 and LP40 stored for 3 and 35 days with 2000 ppm of H₂O (from top to bottom, respectively). (d) ³¹P NMR spectra of fresh LP40 with 2% LiTDI and LP40 stored for 3 and 35 days with 2% LiTDI and 2000 ppm of H₂O (from top to bottom, respectively). Insets are enlarged by the same factor.

seriously challenge the use of LiPF₆ in commercial cells.^{4,7} Lithium 2-trifluoromethyl-4,5-dicyanoimidazole (LiTDI) has been previously studied as a promising candidate to replace LiPF₆ in electrolytes for lithium-ion batteries.^{23,24} Unlike LiPF₆, LiTDI has proven to be highly stable against water and shows promising performance with various electrode materials.^{25–30} However, LiTDI-based electrolytes, especially in the absence of SEI-forming additives, exhibit rather poor compatibility with anode materials, resulting in high irreversible capacities and low Coulombic efficiencies.^{26,31}

In this study, we have applied LiTDI primarily as a moisture scavenger with the aim of suppressing the degradation of the electrolyte. The chemical and thermal stability of the conventional liquid electrolyte LP40 [1:1 (v/v) ethylene carbonate/diethyl carbonate; 1 M LiPF₆] with and without LiTDI additives are compared. The water uptake process of LiTDI is investigated via vibrational spectroscopy and density functional theory (DFT) theoretical calculations. The influence of a small amount of LiTDI additive on the battery performance is demonstrated with LiNi_{1/3}Mn_{1/3}Co_{1/3}O₂ (NMC) half-cells as well as NMC/graphite full cells at an elevated temperature of 55 °C. Moreover, we also perform post-mortem analyses on the interfacial chemistry of the electrode and the stability of the electrolyte.

RESULTS AND DISCUSSION

Chemical Stability of LiPF₆-Based Electrolytes at a High Moisture Level with and without the LiTDI Additive. The moisture content of as-received LP40 is 2.4 ± 0.2 ppm as determined by Karl Fischer titration. Therefore, the degradation of the LP40 electrolyte is expected to be rather slow during storage at room temperature. To visualize the effect of applying LiTDI as a moisture scavenger more efficiently, the water content of the electrolytes, with and without LiTDI, was enriched by 2000 ppm to accelerate electrolyte degradation.

Figure 1a shows the appearance of fresh LP40 and LP40 with 2% LiTDI (~0.1 M, denoted 2% LiTDI), which are colorless and light yellow, respectively. It is worth noting that in the case of the 2% LiTDI electrolyte enriched with 2000 ppm of H₂O, the H₂O:LiTDI molar ratio is ~1:1. After water enrichment and storage for 100 days at room temperature, a distinct difference in color change was observed. The LP40 electrolyte turned dark orange, while the 2% LiTDI electrolyte remained light yellow, as shown in Figure 1b. This is consistent with the observation reported by Wilken et al. that the color of the electrolyte (1:2 EC:DMC weight ratio, 1 M LiPF₆) changed from colorless to brown after being stored at 85 °C for 8 days.¹¹ The authors revealed that multiple LiPF₆ decomposition products, such as POF₃, OPF₂OR, etc., were generated during storage at 85 °C based on nuclear magnetic resonance spectroscopy (NMR) characterizations. Therefore, the color change could be used as an indication of the degradation of the LiPF₆ electrolyte.

¹⁹F and ³¹P liquid NMR characterizations were performed on anhydrous LP40, 2% LiTDI, and their H₂O-enriched equivalents after certain storage times. The ³¹P spectra are shown in panels c and d of Figure 1. The ³¹P spectrum of the fresh LP40 electrolyte contains only a strong septuplet signal centered at −143.5 ppm (sept, ¹J_{P-F} = 711 Hz), which is attributed to the PF₆[−] anion. The corresponding ¹⁹F signal is found at a chemical shift of 70 ppm as shown in Figure S1. After H₂O enrichment and storage for 3 days at room temperature, a new triplet signal centered at −15.3 ppm (¹J_{P-F} = 949 Hz) appeared. This feature is attributed to OPF₂OR type compounds (5% of the total phosphorus-containing species according to NMR quantification), most likely OPF₂(OEt) because of the presence of DEC in the electrolyte. This is a commonly observed degradation product of LiPF₆-based electrolytes.^{9–11,32} Furthermore, as the storage time progresses to 35 days, a new doublet peak centered at −7.2 ppm (¹J_{P-F} =

900 Hz) matching OPF(OR)_2 type species appeared, and the total amount of degradation products increased to 25%.

Interestingly, with the addition of 2 wt % LiTDI to the LP40 electrolyte, no such signals of the LiPF_6 hydrolysis species are visible after storage for 3 or even 35 days after the water enrichment. All three samples that contained 2 wt % LiTDI showed only the PF_6^- signal in the ^{31}P spectra, as shown in Figure 1d. These NMR results fit very well to the observation that there was no significant change in the color of the electrolyte after a long storage time.

The LP40 electrolytes, with and without 2% LiTDI (not water-enriched), were stored at 80 °C for 4 days for thermal stability examination, and the corresponding ^{31}P NMR spectra are shown in Figure 2. Multiple LiPF_6 decomposition products

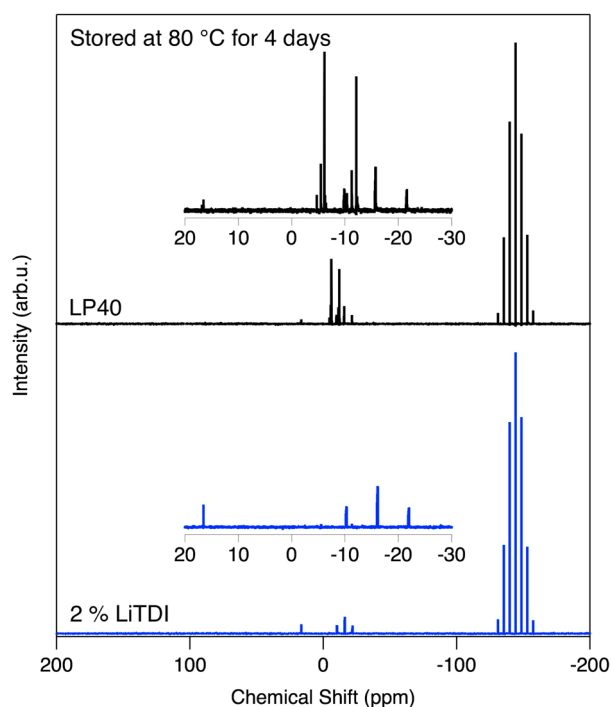


Figure 2. Electrolyte thermal stability. ^{31}P spectra of the LP40 and LP40/2% LiTDI electrolytes after storage for 4 days at 80 °C. Insets are enlarged by the same factor.

were found in both electrolytes. Detailed signal assignment of the degradation products for the LP40 electrolyte (Figure S2) shows that the products are OPF_2OR , multiple OPF(OR)_2 type compounds, and OP(OR)_3 . Quantification analysis reveals the relative amounts of decomposition products and PF_6^- are 42 and 58%, respectively. In the presence of 2% LiTDI, fewer species are formed after storage and the amount of degradation products decreases to 29%, indicating a much improved thermal stability caused by introduction of the LiTDI additive.

Both aging studies, i.e., at high moisture levels and at elevated temperatures, clearly show that the stability of the electrolyte is substantially enhanced in the presence of the LiTDI additive. These results indicate that LiTDI plays a key role in achieving the higher stability, especially in the electrolyte that contains a high level of moisture. Previously, Scheers et al. concluded that LiTDI is chemically stable against water according to Raman investigations on the solid LiTDI salt and 1 M LiTDI aqueous solution.²⁵ Thus, it is highly unlikely that this improvement in stability is achieved by consuming the H_2O via chemical

reactions between LiTDI and H_2O . Therefore, we propose that the LiTDI rather serves as a moisture scavenger that can efficiently absorb water. With smaller amounts of free water molecules available in the electrolyte, LiPF_6 hydrolysis is effectively suppressed as a result.

Water Uptake Process. Niedzicki et al. reported a large fraction (>80%) of LiTDI ion pairs in a PC solution at concentrations between 0.01 and 1 M.³³ Therefore, absorption of water by LiTDI in the electrolyte system investigated here is likely to be rather similar to its behavior in the solid state. Air exposure experiments were performed on dehydrated LiTDI solids, and Fourier transform infrared spectroscopy (FTIR) was used to trace the evolution of characteristic vibrational modes after different air exposure times (relative humidity of ~75%). Three important regions, which are attributed to the nitrile stretching mode [$\nu(\text{C}\equiv\text{N})$], H_2O bending mode [$\nu_2(\text{H}-\text{O}-\text{H})$], and $\text{N}-\text{C}-\text{N}$ bending mode [in the imidazole ring, $\nu(\text{N}-\text{C}-\text{N})$], in the FTIR spectra of the dehydrated and air-exposed LiTDI are shown in Figure 3a (the full range of spectra is presented in Figure S3). No pronounced signal can be observed at the water bending region of the dried LiTDI sample, or at the $\text{O}-\text{H}$ stretch region (Figure S3), indicating that the LiTDI has been sufficiently dehydrated. This is also confirmed by the TGA results that show no significant weight loss before 220 °C, at which temperature the LiTDI starts to decompose, as shown in Figure 3b. For the anhydrous LiTDI solid, the $\text{C}\equiv\text{N}$ stretching and the $\text{N}-\text{C}-\text{N}$ bending signals are located at 2266 and 996 cm^{-1} , respectively.

Once the anhydrous LiTDI solid is exposed to air, a H_2O bending signal at 1635 cm^{-1} appears, indicating uptake of H_2O by LiTDI. In the meantime, this hydration process causes peak shifts in both the nitrile (to lower wavenumbers) and $\text{N}-\text{C}-\text{N}$ vibrational (to higher wavenumbers) modes. After exposure for 40 min, a different water uptake process starts to emerge. New signals at 2244 and 1658 cm^{-1} appear in $\text{C}\equiv\text{N}$ stretching and H_2O bending modes, respectively, while no noticeable change is observed from the nitrogen ring-related bending signal. The results suggest that there are at least two distinct water absorption (WA) processes for the LiTDI solid. The first process (WA-1) occurs during the first 20 min of air exposure, and both the nitrile and the imidazole ring groups are involved in water absorption; the second process (WA-2) starts later but before 40 min and involves mainly the nitrile groups. The spectra obtained after exposure for 60 min, 120 min and 5 days show no significant difference, suggesting that the water uptake process of LiTDI is rather fast.

TGA revealed that there are also two distinct water removal (WR) steps from the hydrated LiTDI solid. One occurs below 75 °C (WR-1, 14.5 wt % loss) and the other between 75 and 140 °C (WR-2, 7.2 wt % loss), as shown in Figure 3b. Interestingly, it is worth noting that the molar ratio between these two types of absorbed water and the LiTDI is close to 2:1:1, indicating that the two LiTDI hydrates are monohydrate $\text{LiTDI}\cdot\text{H}_2\text{O}$ and trihydrate $\text{LiTDI}\cdot 3\text{H}_2\text{O}$. In addition, the water molecule in $\text{LiTDI}\cdot\text{H}_2\text{O}$ is more strongly coordinated to the LiTDI, because the temperature required to have it removed is higher, compared to that of the $\text{LiTDI}\cdot 3\text{H}_2\text{O}$ complex. $\text{LiTDI}\cdot\text{H}_2\text{O}$ was successfully obtained by drying the long time air-exposed LiTDI ($\text{LiTDI}\cdot 3\text{H}_2\text{O}$) under vacuum at 80 °C for 2 h, supported by the TGA result (Figure 3b, green curve). The corresponding IR spectrum is presented in Figure 3a (green curve, $\text{LiTDI}\cdot\text{H}_2\text{O}$). This spectrum is found to be highly similar to that of the 5 min air-exposed LiTDI. This reveals that the

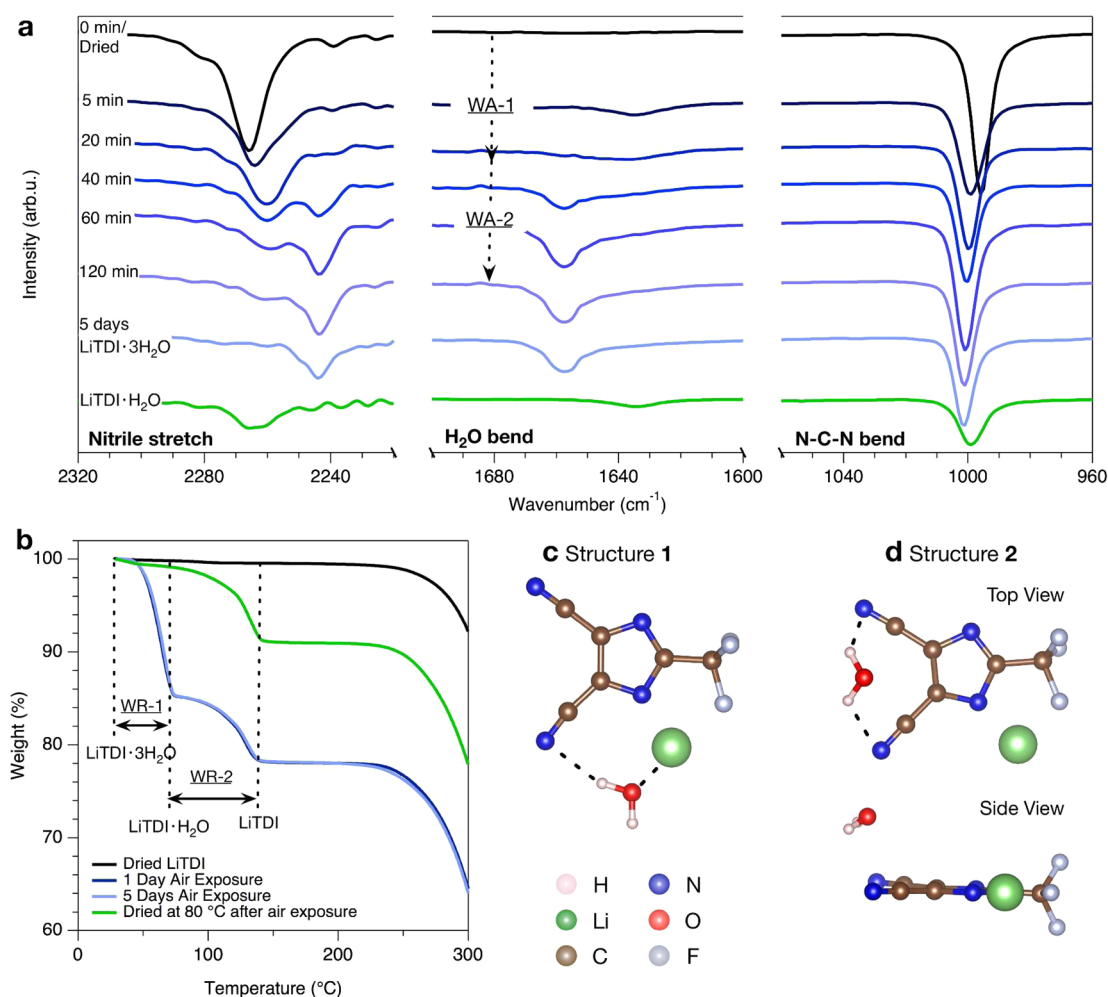


Figure 3. LiTDI water uptake. (a and b) FTIR spectra and TGA results of dehydrated LiTDI solid, air-exposed LiTDI at different exposure times, and a LiTDI sample dried at 80 °C after air exposure, respectively. WA and WR stand for water absorption and water removal, respectively. (c and d) Structures of LiTDI·H₂O obtained from DFT calculations.

LiTDI monohydrate is formed during the WA-1 process. The LiTDI trihydrate is then accordingly produced during the WA-2 process.

To gain insight into the favored coordination between water molecules and LiTDI, DFT calculations were performed, both under vacuum and with an implicit solvent (ethylene carbonate; EC considered the solvent) model. For the LiTDI salt, the most stable configuration is that in which the Li ion is connected to one of the N atoms in the ring and a fluorine from the CF₃ group. This is in agreement with a previous study reported by Scheers et al.²⁵ Four possible structures were considered for interaction between a H₂O molecule and the LiTDI salt in a 1:1 ratio, and the optimized structures are shown in Figure S4. The corresponding binding energies (calculated on the basis of eq 4) are listed in Table S1.

$$E_{\text{bind}} = E_{\text{H}_2\text{O-LiTDI}} - (E_{\text{H}_2\text{O}} + E_{\text{LiTDI}}) \quad (4)$$

Among these possibilities, structure 1 (Figure 3c) shows the lowest binding energy in both gas phase and implicit solvent modes (−29.48 and −9.46 kcal mol^{−1}, respectively), meaning that this is the most stable structure. In this structure, the oxygen in the water molecule is coordinated to the Li cation, and one of the hydrogen atoms is bonded to the nitrogen in the nitrile group via hydrogen bonding. This finding is in very good

agreement with the evolution of IR spectra during the WA-1 process, i.e., formation of LiTDI·H₂O, which shows a shift in wavenumber to lower values for the C≡N stretching signal and to higher values for the N–C–N bending signal. For the case of LiTDI·3H₂O, we proposed above that only the nitrile groups are involved in the formation process because no significant evolution is observed apart from the nitrile stretching mode. Structure 2 (Figure 3d) indeed shows the possibility of such a water coordination condition: the H₂O molecule bridges the two nitrile groups via hydrogen bonding. It should be noted that such structures are not the exact structures of LiTDI monohydrate or trihydrate. Instead, these are proposed as preferred coordinations between LiTDI and H₂O, based on spectroscopic observation and DFT calculation. This also presents an opportunity to study the LiTDI hydrate structures and LiTDI–H₂O interaction in the electrolyte comprehensively for the future.

To briefly summarize, we propose that LiTDI·H₂O is first formed when anhydrous LiTDI is exposed to air, and LiTDI·3H₂O emerges as the exposure progresses. Structure 1 is the most favored LiTDI·H₂O coordination, as indicated by both experimental IR and TGA results, as well as vacuum gas phase and implicit solvent (EC) model DFT calculations.

Performance of NMC/Li Cells at an Elevated Temperature of 55 °C. The moisture-scavenging effect of LiTDI on

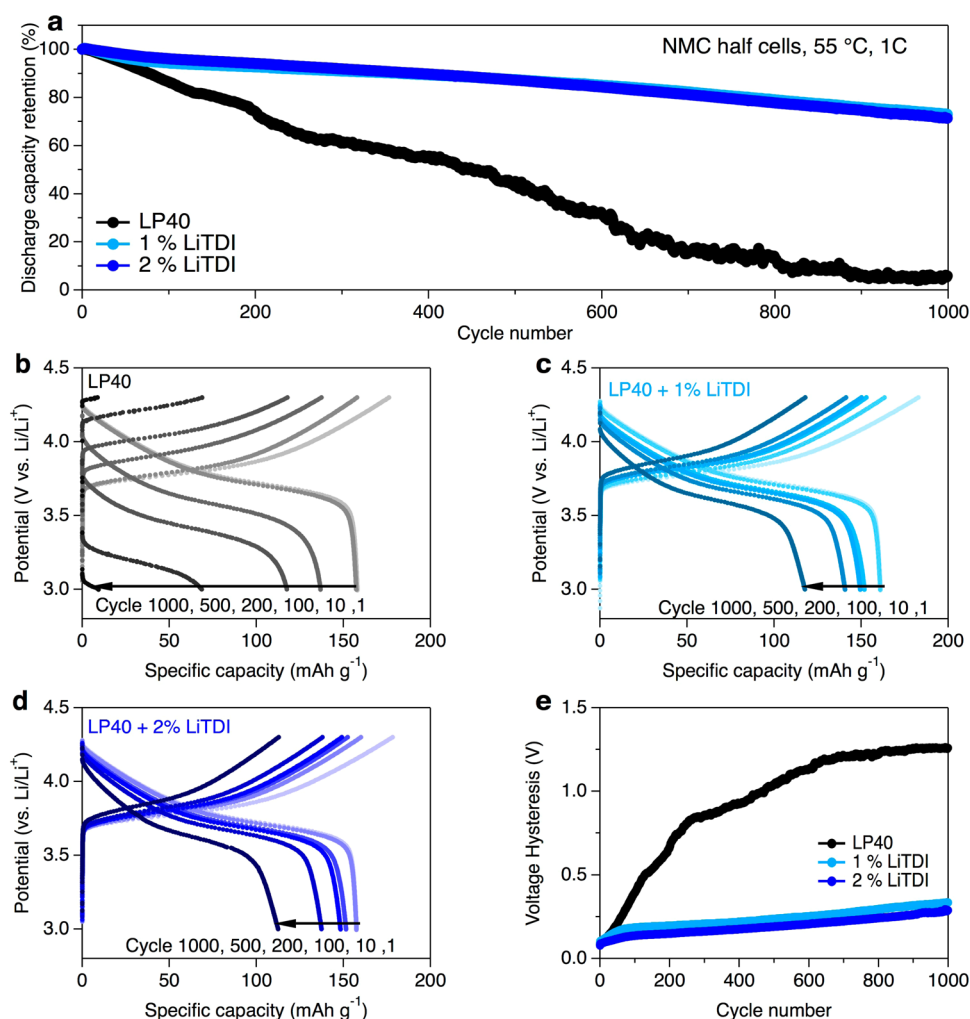


Figure 4. Galvanostatic cycling performance of NMC/Li cells at an elevated temperature of 55 °C; 1 C charging and discharge rate. (a) Discharge capacity retention of the NMC half-cells using LP40, 1% LiTDI, and 2% LiTDI electrolytes. (b–d) Voltage profiles of the NMC half-cells using LP40, 1% LiTDI, and 2% LiTDI electrolytes, respectively. Cycles 1, 10, 100, 200, 500, and 1000 of each cell are included. (e) Voltage hysteresis profile of the LP40, 1% LiTDI, and 2% LiTDI cells.

battery performance is investigated using an NMC half-cell configuration at an elevated temperature of 55 °C and 1 C charging and discharging rates. The electrolytes are as-received LP40, LP40 with 1% LiTDI, and LP40 with 2% LiTDI, and the obtained electrochemical performances are presented in Figure 4. Figure 4a shows the discharge capacity retention of the NMC half-cells using the three investigated electrolytes. The LP40 cell shows a continuous and fast capacity decay. By the 140th cycle, the discharge capacity of the cell decreased to 80% of the initial cycle value ($\sim 160 \text{ mAh g}^{-1}$, calculated on NMC active material). With the introduction of the LiTDI additive to the LP40 electrolyte, capacity retention was substantially enhanced. With a much slower capacity decay, the 80% initial capacity retention threshold is extended to the 830th and 750th cycles, with 1% LiTDI and 2% LiTDI additive electrolytes, respectively. These differences can be considered to be within the margin of error, and it is interesting to note that >1% amounts of LiTDI additive do not seem to affect the cycling stability. Moreover, without the LiTDI additive, both charge and discharge voltage profiles changed dramatically over the long-term battery cycling, showing a continual increase in voltage hysteresis. The voltage slope of the charging process is gradually shifting to higher values, whereas the discharging

slope is shifting to lower values as the cycling progresses (Figure 4b). In the presence of small amounts of LiTDI in the electrolyte, both average charge and discharge voltages are extremely stable over the long-term cycling, and the capacity retention performance is therefore much improved (Figure 4c,d). The evolution of cell voltage hysteresis is plotted in Figure 4e. The voltage hysteresis of the LP40 cell rapidly increased from 0.08 to >1 V after 500 cycles, while the corresponding values of the LiTDI-containing cells remain as low as $\sim 0.25 \text{ V}$.

The galvanostatic cycling results show that a small amount of LiTDI additive, 1 wt %, is sufficient to substantially enhance the performance of NMC/Li cells at an evaluated temperature of 55 °C. This is primarily attributed to the effectively limited cell voltage hysteresis increase during long-term cycling.

Post-Mortem Analysis. Given the dramatic difference in the evolution of cell voltage hysteresis, it is important to understand the development of cell impedance during cell cycling. EIS results of NMC half-cells with LP40, 1% LiTDI, and 2% LiTDI after 250 cycles are shown in Figure 5a in the form of a Nyquist plot. It should be noted that these tests were performed at room temperature because of the poor quality of the data at 55 °C, especially at low frequencies. This is probably

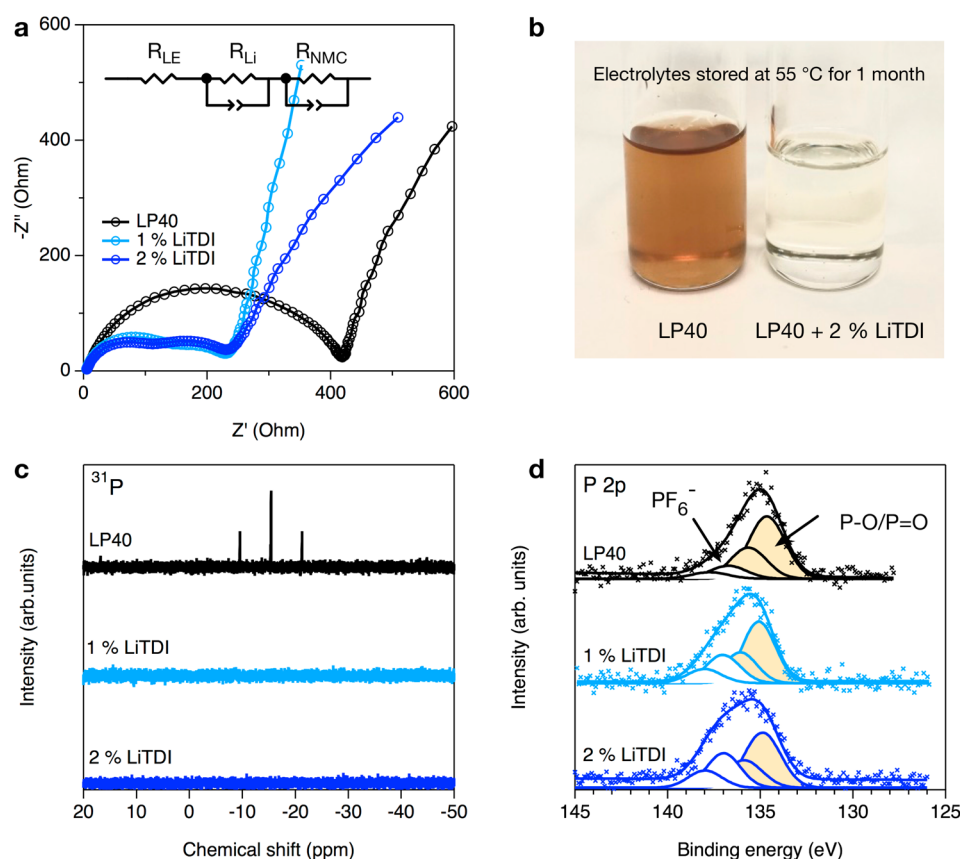


Figure 5. Electrolyte aging and postcycling analyses. (a) EIS results of the NMC/Li cells after 250 cycles at 55 °C. (b) Images of the LP40 and 2% LiTDI electrolytes after storage for 1 month at 55 °C. (c) ^{31}P NMR spectra of the LP40, 1% LiTDI, and 2% LiTDI electrolytes recovered from the NMC/Li half-cells after 250 cycles at 55 °C. (d) P 2p XPS spectra of the NMC electrodes from the NMC/Li half-cells with LP40, 1% LiTDI, and 2% LiTDI electrolytes after 250 cycles at 55 °C.

due to the instability of the system, for example, thermal vibrations. Still, a rather large difference can be seen between the LP40 cell and the other two cells that contain LiTDI in the electrolyte. All of the three spectra, especially in the cases of 1% LiTDI and 2% LiTDI cells, show two distinct semicircles at two frequencies (Bode representations in Figure S5). These two semicircles are attributed to the two electrode–electrolyte interfaces in batteries, i.e., NMC–electrolyte and Li–electrolyte. To identify the contributions, EIS experiments were also performed on three-electrode cell setups using NMC as the working electrode (WE) and Li as the reference electrode (RE) and counter electrode (CE). Cells with LP40 and 1% LiTDI were galvanostatically cycled for 50 cycles at a C-rate of 1C at 55 °C prior to EIS characterization. The obtained results for the WE, as well as cell impedance (WE/CE), are presented in Figure S6. The spectrum of WE/CE shows that the profile of $|Z|$ below 100 Hz is highly similar to that in the same region of the WE, suggesting that the WE, i.e., NMC electrode, dominates the contribution of impedance to the cell in the low-frequency range. On this basis, we assign the semicircle at a lower frequency to the NMC–electrolyte interface, and accordingly, the other semicircle at a higher frequency is attributed to the Li–electrolyte interface.

Equivalent circuit fitting was performed on the EIS results of the NMC/Li cells after 250 cycles between the frequencies of 200 kHz and 10.9 Hz (the semicircle region) with the model presented in Figure 5a, and the fitting parameters are summarized in Table 1. In the equivalent circuit, there are

Table 1. Fitted Results of the EIS Spectra between 100 kHz and 10.9 Hz

	LP40	1% LiTDI	2% LiTDI
R_{LE} (Ω)	4.4	3.6	4.1
$R_{ct,Li}$ (Ω)	103.2	96.1	72.6
$R_{ct,NMC}$ (Ω)	322.6	163.2	195.8
sum (Ω)	430.2	262.9	272.5

primarily three components that contribute to the cell resistance: ohmic resistance of the liquid electrolyte (R_{LE}), charge transfer resistance at the lithium electrode ($R_{ct,Li}$), and charge transfer resistance at the NMC electrode ($R_{ct,NMC}$). It can be seen that there is no dramatic difference in R_{LE} and $R_{ct,Li}$ among the three cells. However, the charge transfer resistance at the NMC electrode ($R_{ct,NMC}$) of the LP40 cell (322.6 Ω) is substantially larger than in the 1% LiTDI and 2% LiTDI cells (163.2 and 195.8 Ω , respectively). As a result, the overall cell resistance (the sum of R_{LE} , $R_{ct,Li}$, and $R_{ct,NMC}$) of the LP40 cell is also much higher than that of the cells that contain the LiTDI additive.

These results are in good agreement with the galvanostatic cycling results presented in Figure 4, which shows that the cell voltage hysteresis of the LP40 cell is substantially higher than for the LiTDI-containing cells. Furthermore, this is primarily attributed to the higher charge transfer resistance at the NMC electrode.

It is widely acknowledged that degradation of LiPF_6 , especially at elevated temperatures, will have other severe

detrimental effects on the battery, for example, overgrowth of the SEI, which in turn results in an increase in cell impedance.^{16,18,19} We demonstrated above that the chemical and thermal stability of the LP40 electrolyte during storage can be improved with the moisture scavenger LiTDI. Figure 5b shows an image of the LP40 and 2% LiTDI electrolytes that were stored at 55 °C for 1 month. The dark brown color of the aged LP40 electrolyte indicates a more severe degradation compared to that of the 2% LiTDI electrolyte. Here, we also investigated the electrolyte stability in the operating cells after 250 cycles at 55 °C. ³¹P NMR spectra of the recovered LP40, 1% LiTDI, and 2% LiTDI electrolytes from cycled batteries are presented in Figure 5c. The figure is magnified to a shift range of 20 to −50 ppm, where the LiPF₆ degradation products are usually found (spectra with the full chemical shift range are shown in Figure S7). OPF₂(OR) type species (triplet signal entered at −15.4 ppm) are found to be the main LiPF₆ decomposition products in the LP40 electrolyte after 250 cycles at 55 °C. On the other hand, in the presence of 1 and 2% LiTDI, there is no detectable degradation of LiPF₆ from the recovered electrolyte. Moreover, as revealed by XPS (Figure 5d), the degradation of LiPF₆ is also suppressed at the electrode surface when using the LiTDI additive. The P 2p spectra of the NMC electrodes cycled with LP40, 1% LiTDI, and 2% LiTDI show two doublet signals, which have the 2p_{3/2} signals located at ~136.5 and ~134.5 eV, and correspond to PF₆[−] and P–O/P = O phosphate compounds, respectively.^{34,35} It can be noticed from these results that, in the presence of the LiTDI additive, the amount of phosphorus-containing degradation products (with respect to PF₆[−] residue) is smaller in both cases than for the LP40 sample.

The post-mortem analysis results thus strongly indicate that the stability of the LiPF₆ electrolyte is significantly improved also during battery operation at elevated temperatures, resulting in a substantially enhanced cycling performance.

NMC/Graphite Full Cells. To demonstrate the effects of the LiTDI additive in conventional full cells for its potentially practical use, NMC/graphite full cells with and without the LiTDI additive were tested, and the battery cycling performance is presented in Figure 6.

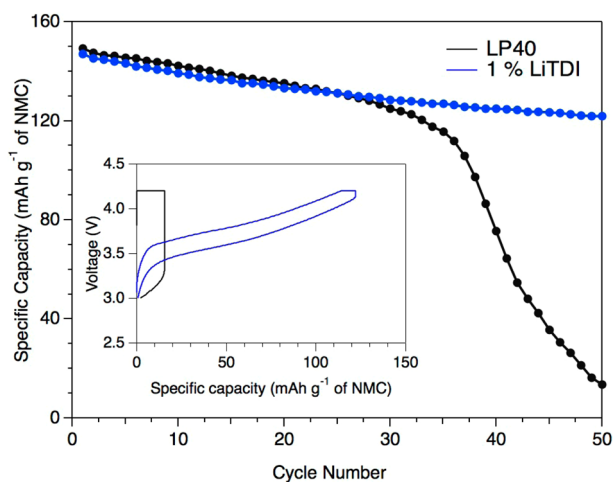


Figure 6. Cycling performance of NMC/graphite full cells with the LP40 electrolyte and LP40/1% LiTDI electrolyte. The inset shows voltage profiles of the full cells at the 50th cycle.

The NMC/graphite cell using the conventional LP40 electrolyte exhibited a steady decrease in capacity during the first 25 cycles, from which the capacity dropped very rapidly to ~20 mAh g^{−1} by the 50th cycle. The corresponding voltage profile shows that charge capacity was dominated by the contribution from the constant voltage step by the end of the cycling, suggesting a high cell impedance and severe kinetic hindrance. However, in the presence of 1% LiTDI in the electrolyte, the retention of capacity was improved and kept above 120 mAh g^{−1} by the end of cycling. These full cell results further confirm that LiTDI can be used as a promising electrolyte additive in lithium-ion batteries, especially at elevated temperatures.

CONCLUSIONS

In summary, we demonstrate that the LiTDI salt can be used as a highly efficient moisture-scavenging additive in lithium-ion battery electrolytes. With a small amount of 2 wt % in the commonly used LP40 electrolyte, which has also been enriched with 2000 ppm water, there was still no detectable degradation of the LiPF₆ electrolyte after storage for 35 days. The LiTDI-containing LP40 electrolyte remained light yellow and transparent, indicating that the integrity is much better preserved compared to that of the LP40 electrolyte. Moreover, the thermal stability of the electrolyte is also improved via application of the LiTDI additive. Experimental and theoretical studies revealed that the structure 1, which has the oxygen in the water molecule coordinated to the Li⁺ and one of the hydrogens coordinated to the nitrogen in the nitrile group, is the most favored LiTDI–H₂O coordination. The application of the LiTDI additive to the battery shows a boost in capacity retention performance of NMC/Li cells at an elevated temperature of 55 °C. Without additives, the charge transfer resistance at the NMC electrode dramatically increased as the cycling progressed. Because of the much improved electrolyte stability with the LiTDI additive, the LiTDI-containing NMC/Li cells showed a slow increase in voltage hysteresis and superior galvanostatic cycling performance. The proof-of-concept investigation of NMC/graphite full cells shows that LiTDI is also effective in such a full cell configuration, and its practical use is promising. Finally, it can be envisioned that these findings could potentially contribute to significant energy savings by reducing the necessity of strongly drying all different battery components.

METHODS

Battery Preparation. LiTDI was provided by the Polymer Ionics Research Group at Warsaw University of Technology, and the synthesis was reported elsewhere.²⁴ LP40 [1 M LiPF₆, 1:1 (v/v) EC:DEC, BASF] was used as the control electrolyte, and LiTDI-containing electrolytes were prepared by mixing dried LiTDI with LP40 in a weight ratio. For the investigation of chemical stability with and without water enrichment, 2000 ppm (weight) of deionized water was added to LP40 and to the LP40 electrolyte with a 2% LiTDI additive (denoted 2% LiTDI), respectively. Water enrichment was performed in an argon-filled glovebag to minimize H₂O contamination from air. Composite LiNi_{1/3}Mn_{1/3}Co_{1/3}O₂ electrodes were fabricated using the NMC material (NMC, MTI Corp.), C65 Conductive Carbon Black (Timical), and poly(vinylidene difluoride-co-hexafluoropropylene) (PVdF-HFP, Kynar FLEX 2801, Arkema) in an 80:10:10 mass ratio. N-Methylpyrrolidone (NMP, Aldrich) was used as a solvent, and the slurry was ball-milled for 1 h and casted on a carbon-coated aluminum foil. The loading of active material for NMC is ~3 mg cm^{−2}. Prior to battery assembly, NMC electrodes were vacuum-

dried at 120 °C for 12 h. NMC electrodes used for NMC/graphite full cells were prepared using the same approach, and the loading of NMC was $\sim 11 \text{ mg cm}^{-2}$. Graphite electrodes (active mass loading of $\sim 5 \text{ mg cm}^{-2}$) were comprised of 94 wt % SMG-A3 graphite (Hitachi), 2 wt % SuperP (Imerys), 2 wt % carboxymethyl cellulose sodium salt (CMC-Na), and 2 wt % styrene-butadiene rubber (SBR, Styron). The slurry was prepared on a 1 L scale, employing a cross-arm and butterfly stirrer couple. The dry powders were first blended prior to the gradual addition of deionized water. After being stirred for 2 h, the slurry was degassed and stored under reduced pressure overnight. The following day, the slurry was stirred again for 30 min and then applied to copper foil (Schlenk Metalfolien, 12 μm) on a pilot coater. The electrode coating was calendered to a density of 1.4 g mL^{-1} .

The batteries investigated in the study were all assembled as pouch cells. The NMC/Li half-cell consisted of an NMC working electrode, a lithium metal counter electrode, and electrolyte-soaked separators. It should be noted that one piece of Solupor and glass fiber separators were used for each half-cell, with the Solupor facing the working electrode to ensure a clean (fiber-free) surface for further characterization. The NMC/graphite full cell consisted of an NMC working electrode, a graphite electrode, and one Celgard 2325 separator. The amount of added electrolyte was 50 μL for each cell.

Characterizations. Galvanostatic cycling of the NMC half-cells at an elevated temperature of 55 °C was performed on HPC Novonix high-precision cycling equipment at the same charge and discharge rate of 1C. Cycling on NMC/graphite full cells was first performed with two cycles at a rate of C/10 (based on the active NMC material) between 3.0 and 4.2 V, followed by a C/2 cycling. An additional constant voltage step was added to the charging process, and cells were kept at 4.2 V until the current was lower than C/20. Electrochemical impedance spectroscopy (EIS) measurements were performed on a VMP potentiostat with a frequency range of 200 kHz to 100 mHz at an amplitude perturbation of 10 mV, and the tested batteries were cooled to room temperature prior to measurements.

The moisture level in the electrolyte samples was characterized by the Karl Fisher titration method using a 756 KF Coulometer (Metrohm). All titration samples were tested under three parallel measurements. Thermogravimetric analyses were performed on a TA Instruments TGA Q500 device at a heating rate of 5 °C/min under a nitrogen flow. FTIR measurements were taken on a PerkinElmer attenuated total reflectance-Fourier transform infrared (ATR-FTIR) spectrometer in the frequency range of 650–4000 cm^{-1} . Nuclear magnetic resonance (NMR) spectra were recorded on a 400 MHz Varian Unity spectrometer, and all NMR measurements were taken using a DMSO- d_6 solvent. The electrolyte samples from cycled batteries were recovered by rinsing the separator in the DMSO- d_6 solvent. X-ray photoelectron spectroscopy (XPS) experiments were performed on a PHI 5500 system with monochromatized Al K α radiation ($h\nu = 1486.6 \text{ eV}$) as the light source. The samples were carefully rinsed with dimethyl carbonate (DMC) in an argon-filled glovebox and then transferred to the spectrometer with an airtight transfer chamber to prevent any contamination with air. The obtained spectra were curve fitted with Igor software (version 6.36) using mixtures of Gaussian and Lorentzian line shapes (the Lorentzian full width at half-maximum was kept below 0.1 eV). The P 2p spin-orbit splitting energy value used in this work was 1.1 eV. All spectra were calibrated with respect to the C 1s peak of hydrocarbon species at 285 eV.

DFT Calculations. DFT calculations have been conducted using the B3LYP hybrid XC-functional^{36–39} and 6-311+G* basis set^{40–42} with the Gaussian G09 package.⁴³ The Grimme-D3 dispersion correction has been used to include long-range interactions.⁴⁴ Furthermore, the vibrational frequencies are also calculated for the optimized structure in the vacuum, and no imaginary frequency is observed, indicating the structures are stable minima on the potential energy surface.

DFT calculations were also performed using an implicit solvent model. The single-point calculations based on the optimized structures in vacuum were performed using the polarizable conductor calculation model (CPCM).^{45,46} The solvent model in this study is ethylene

carbonate, and the parameter for the CPCM calculations is the dielectric constant of EC, 89.78 at 298.15 K.⁷

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.chemmater.6b05247.

Figures S1–S7 and Table S1 (PDF)

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Notes

The authors declare no competing financial interest.

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